

Course Name: Computer Graphics Fundamentals

Project Title: Practical Work

Team Members: Yernur Amanturlly IT-2508

Instructor: Sabina Saleshova

Date of Submission: 23.09.2026

Table of Contents.

1. Introduction.

I created Night landscape, there is mountain, day/night switch mode and stars using modern technologies OpenGL, glad, etc.

2.Literature Review.

An open image is assembled from dots, lines, and triangles. The data is transferred to the video card through the VBO and VAO buggers, the eiders are responsible for displaying the information.

3.Project Description.

This project has created a simple 2D image of a night landscape. GL*Point, GLLines, GL_triangles and space event*.

4.Technical implementation.

Instruments: C++, OpenGL 3.3, GLFW, GLAD.

5.Graphics and Rendering

The sky, the earth and the mountain are drawn in triangles, the moon and the stars.

The sky gradient is made by automatically mixing vertex colors in a fragment shader.

The Space key changes the value of the isDayMode uniform variable and switches day/night.

6.Testing and Evaluation

The program is working stably.
GL_PROGRAM_POINT_SIZE function is enabled to display a large moon.

7.Conclusion

This task is easy, but I have a little bit problem with switch mode, I didn’t how to realize it, and so I had to read documentations.